Faithful group actions may lose group-algebra information

WRITTEN PROOF + EXACT EXAMPLE / standard representation of S_4

$$V = \{z \in \mathbb{Q}^4 : z_1 + z_2 + z_3 + z_4 = 0\}$$

$$H = \langle (1234) \rangle$$
$$E_H|_V = 0$$

$$K = \langle (12) \rangle$$
$$E_H E_K = E_K E_H = 0$$

$$H \not\lhd S_4 \qquad HK \neq KH$$

The action on V is faithful. The induced group-algebra action is not.